This set of problems is a compilation of all questions from the Spring 2022 and Spring 2023 Large Sample Theory course homework at UF.

Large Sample Theory Practice Problems

1 Probability Definitions

Suppose $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}$, the Borel σ-algebra, and $P = \lambda$, the Lebesque measure, also known as the length measure (i.e., $\lambda([a, b]) = b - a$, for $0 \leq a \leq b \leq 1$). Suppose X is a random variable on the probability space (Ω, $\mathcal{F}$, P) defined as

$$X(\omega) = \omega, \forall\, \omega \in \Omega$$

(i) What is F_x in this case? What is the name of this distribution?

(ii) Compute $E(X)$ in two ways and show that they are equal. That is

$$\int X dP = \int_0^1 x dF_X(x).$$

2 Modes of Convergence

1. (Prove or provide a counter example) If $X_n \xrightarrow{p} c$, then $X_n \xrightarrow{L_1} c$.

2. (Prove or provide a counter example) If $X_n \xrightarrow{L_1} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{L_1} X + c$.

3. (Prove or provide a counter example) If $X_n \xrightarrow{L_1} X$ and $Y_n \xrightarrow{p} c$ and $|Y_n| \leq 1\ \forall n$ with probability one, then $X_n + Y_n \xrightarrow{L_1} X + c$.

4. (Prove or provide a counter example) If $E(X_n - Y_n)^2 = O(n^{-2})$ and $\sqrt{n}(X_n - c) \xrightarrow{d} N(0, 5)$, then $Y_n \sim AN(c, 5/n)$.

5. Suppose X_n are defined on the same probability space and that

$$\sum_{i=1}^{\infty} E(|X_n|^p) < \infty, \text{ for some } p > 0.$$

   (a) Show that $X_n \to 0$ completely.

   (b) Is it true that $X_n \to 0$ in probability?

   (c) Is it true that $X_n \to 0$ in L_1, if $p \geq 1$?

3 Delta Method and Influence Functions

Let $0 < p < 1$, $\xi_p = F^{-1}(p)$, be the population quantile where we assume that F is differentiable at ξ_p with $F'(\xi_p) > 0$. In fact, assume $F' = f$ where f is the Lebesgue density of F. As stated in class

$$\hat{\xi}_p - \xi_p = -\frac{F_n(\xi_p) - p}{f(\xi_p)} + o_p(n^{-1/2})$$

where $\hat{\xi}_p = \inf\{t \ : \ F_n(t) \geq p\}$ is the sample quantile.

1. Show that

$$\sqrt{n}\{\hat{\sigma}_n - s\} = o_p(1),$$

   where

$$\hat{\sigma}_n = \sqrt{\frac{1}{n}\sum(X_i - \overline{X})^2} \quad s = \sqrt{\frac{1}{n-1}\sum(X_i - \overline{X})^2}$$

2. Work out the joint asymptotic distribution of $(s, \hat{\xi}_{3/4} - \hat{\xi}_{1/4})$.

3. Work on your answer when $F = N(\mu, \sigma^2)$.

4. Suppose (X_i, Y_i) are i.i.d. pairs from a bivariate normal distribution such that $corr(X_i, Y_i) = \rho$. Find the asymptotic distribution of $\tanh^{-1}(\hat{\rho})$, where

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \overline{X})Y_i}{\sqrt{\left(\sum_{i=1}^n (X_i - \overline{X})^2\right)\left(\sum_{i=1}^n (Y_i - \overline{Y})^2\right)}}$$

4 MLE and Estimating Equations

1. Consider solving the likelihood score equation $S_n(\theta) = 0$. Suppose $\hat{\theta}_c$ is any consistent estimator. The 1-step MLE is defined as

$$\hat{\theta}_{1,MLE} = \hat{\theta}_c - \frac{S_n(\hat{\theta}_c)}{S_n'(\hat{\theta}_c)}$$

   Show that $\hat{\theta}_{1,MLE} \sim AN(\theta_0, n^{-1}I^{-1}(\theta_0))$. In other words, the 1-step MLE has the same asymptotic distribution as the MLE.

2. Consider a location family $f_\theta(x) = f(x - \theta)$, $\theta \in \mathbb{R}$, where f is a given density which is symmetric around 0. Consider estimating θ by solving an estimating equation of the for

$$\sum_{i=1}^{\infty} \psi(X_i - \theta) = 0.$$

   (a) What should we assume about ψ so that the above estimating equation will lead to an unbiased estimator.

   (b) Consider two such estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ corresponding to two different ψ, say, ψ_1 and ψ_2. Find the joint asymptotic distribution of $(\hat{\theta}_1, \hat{\theta}_2)$. [hint: work with influence functions]

3. Consider a simple linear regression model with zero intercept: $Y_i = \beta x_i + \epsilon_i$, where x_i are observed covariates and ϵ_i are i.i.d. error terms. Let's assume that ϵ_i have a symmetric distribution.

   Consider estimating the regression parameter β using an estimating equation of the form

$$\sum_{i=1}^{n} \psi(Y_i - \beta x_i) x_i = 0.$$

   i. Show that the estimating function used here is unbiased (for zero) under a simple condition on ψ (state the condition)

   ii. Find the asymptotic distribution of $\hat{\beta}$ including variance estimation.

   iii. Specialize to the case of the least squares estimator of β.

5 U-statistic and V-statistic

1. Find the U-statistics U_1 and U_2 for σ^2 and μ_3, respectively. Obtain the joint asymptotic distribution of (U_1, U_2).

2. Consider two independent samples $\{X_1, \ldots, X_n\}$ and $\{Y_1, \ldots, Y_m\}$ distributed F and G respectively (assume both F and G are continuous). Work out the asymptotic null distribution of the test statistic

$$\frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} U(X_i < Y_j)$$

   under $H_0 : F = G$.

3. Support h_1 and h_2 are two kernels of orders p and q respectively. Consider two independent samples $\{X_1, \ldots, X_p\}$ and $\{Y_1, \ldots, Y_q\}$ distributed F and G, respectively. Let $E_F h^1(X_1, \ldots, X_p) = \eta$ and $E_G h^2(Y_1, \ldots, Y_q) = \kappa$.

   (a) Obtain the asymptotic distribution of a two-sample U statistic with kernel

$$\mathcal{H} = (X_1, \ldots, X_p; Y_1, \ldots, Y_q) = h^1(X_1, \ldots, X_p) \cdot h^2(Y_1, \ldots, Y_q)$$

   (b) Specialize to the case when $\eta = \sigma_F^2$ and $\kappa = \sigma_G^2$.

6 Jackknife and Bootstrap approximation

1. Consider a random sample $\{1, 3, 4\}$ from a population. Suppose we want to estimate the population median by the sample median.

   (a) Compute the jackknife pseudo values.

   (b) Compute a variance estimate of the sample median using the jackknife.